

Norway

SLEIPNER

Volve F

F-11

15/9-F-11

Plan: 15/9-F-11 rev0

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54.90m (MI)
Site:	Volve F	MD Reference:	MI @ 54.90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 rev0	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site		Volve F, 15/9				
Site Position:		Northing:	6,478,563.53 m	Latitude:	58° 26' 29.8068 N	
From:	Map	Easting:	435,050.02 m	Longitude:	1° 53' 14.9285 E	
Position Uncertainty:		0.00 m	Slot Radius:	13.200 in	Grid Convergence:	-0.95 °

Well	F-11					
Well Position	+N/-S	4.65 m	Northing:	6,478,568.18 m	Latitude:	58° 26' 29.9566 N
	+E/-W	-0.93 m	Easting:	435,049.09 m	Longitude:	1° 53' 14.8667 E
Position Uncertainty		0.00 m	Wellhead Depth:	91.00 m	Water Depth:	91.00 m

Wellbore	15/9-F-11				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	MNETICREFERENCE	02.12.2010	-1.68	71.63	50,437

Design	15/9-F-11 rev0				
Audit Notes:	Have hanged wellname according to Statoil standard 28.09.2012				
Version:	Phase:	PLAN	Tie On Depth:	145.90	
Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)	
	145.90	4.65	-0.93	62.38	

Survey Tool Program	Date	01.07.2013			
From (m)	To (m)	Survey (Wellbore)	Tool Name	Description	
145.90	650.00	15/9-F-11 rev0 (15/9-F-11)	NOT_useMWD gyro, Baker Hughes'	Baker Hughes' MWD gyro service (GyroTrak)	
650.00	1,366.00	15/9-F-11 rev0 (15/9-F-11)	Magnetic, IFR, mag-corr	Magnetic Tools (MWD, EMS)	
1,366.00	2,572.00	15/9-F-11 rev0 (15/9-F-11)	Magnetic, IFR, mag-corr	Magnetic Tools (MWD, EMS)	
2,572.00	4,550.76	15/9-F-11 rev0 (15/9-F-11)	Magnetic, IFR, non-mag	Magnetic Tools (MWD, EMS)	

Planned Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
145.90	0.00	0.00	145.90	4.65	-0.93	0.00	0.000	0.000	0.000	
545.90	0.00	0.00	545.90	4.65	-0.93	0.00	0.000	0.000	0.000	
745.91	6.67	290.00	745.46	8.63	-11.85	-7.84	1.000	1.000	0.000	
896.74	12.80	262.40	894.12	9.41	-36.67	-29.47	1.500	1.220	-5.489	
1,077.10	12.80	262.40	1,070.00	4.13	-76.28	-67.01	0.000	0.000	0.000	
1,257.34	27.80	264.55	1,238.56	-2.54	-138.27	-125.03	2.500	2.497	0.358	
1,797.13	27.80	264.55	1,716.03	-26.46	-388.92	-358.21	0.000	0.000	0.000	
2,567.54	37.00	69.60	2,442.00	44.90	-345.90	-287.01	2.500	0.358	6.427	

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Planned Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
2,577.54	37.00	69.60	2,449.99	47.00	-340.26	-281.04	0.000	0.000	0.000	
3,009.41	80.19	69.60	2,669.75	172.46	-2.90	76.04	3.000	3.000	0.000	
3,708.56	80.19	69.60	2,788.91	412.60	642.81	759.50	0.000	0.000	0.000	
4,355.07	14.21	69.76	3,204.30	568.48	1,063.45	1,204.47	3.061	-3.061	0.008	
4,550.76	14.21	69.76	3,394.00	585.10	1,108.53	1,252.12	0.000	0.000	0.000	

Design Targets										
Target Name	Dip Angle (°)	Dip Dir. (°)	TVD (m)	+N/-S (m)	+E/-W (m)	Northing (m)	Easting (m)	Latitude		Longitude
- hit/miss target										
- Shape										
F-15 D N Pilot Top Hugir	0.00	360.00	3,204.30	568.48	1,063.45	6,479,131.80	436,113.10	58° 26' 48.7409 N		1° 54' 19.8979 E
- plan misses target center by 0.27m at 4355.07m MD (3204.30 TVD, 568.73 N, 1063.36 E)										
- Rectangle (sides W100.00 H50.00 D0.00)										

Casing Points					
Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)	
202.00	202.00	30" Conductor	30.000	36.000	
1,362.97	1,332.00	20" casing	20.000	26.000	
2,567.54	2,442.00	14" Casing	14.000	17.500	

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Formations						
Measured Depth (m)	Vertical Depth (m)	Name	Lithology	Dip (°)	Dip Direction (°)	
886.60	884.22	Utsira_Fm_Top		0.00		
1,073.70	1,066.68	Utsira_Fm_Base		0.00		
1,221.90	1,206.80	Skade_Fm_Top		0.00		
1,337.90	1,309.82	Skade_Fm_Base		0.00		
2,135.00	2,040.39	Grid_Fm_Top		0.00		
2,204.00	2,109.01	Grid_Fm_Base		0.00		
2,322.30	2,224.46	Balder_Fm_Top		0.00		
2,384.00	2,282.63	Sele_Fm_Top		0.00		
2,450.20	2,342.78	Lista_Fm_Top		0.00		
2,564.70	2,439.73	Ty_Fm_Top		0.00		
2,761.30	2,576.66	Tor_Fm_Top		0.00		
2,795.10	2,595.03	Ekofisk_Fm_Top		0.00		
3,091.20	2,683.69	Hod_Fm_Top		0.00		
3,811.10	2,815.48	Hidra_Fm_Top		0.00		
3,827.90	2,821.54	Roedby_Fm_Top		0.00		
3,918.60	2,862.14	Aasgard_Fm_Top		0.00		
3,977.90	2,895.51	Draupne_Fm_Top		0.00		
4,110.80	2,987.75	Hugin_Fm._VOLVE_Top		0.00		
4,264.50	3,118.67	Hugin_Fm._VOLVE_Base		0.00		

Checked By: _____ Approved By: _____ Date: _____